TEAM ASTERIX · aBAJA SAEINDIA

POWERTRAIN WORKSHOP 2026

Electrical · Electronics · Embedded Systems · Motors · PCB Design

Duration	7 October 2026 – 12 November 2026 (12 scheduled sessions). Additional venue dates: 14–15 November 2026; session details to be notified.
Schedule	Sessions are scheduled across October and November. Dates and topics are listed in the timetable. Online delivery is specified for 17 and 28 October.
Timing	Session timings have not been provided and will be communicated by the workshop team.
Venue	**E6 – Mechanical Seminar Hall** for all sessions, except 14 and 15 November, which are at **E8 – Robotics Seminar Hall**. Online sessions: 17 and 28 October.
Subjects	Network Analysis, Electronic Devices, Microcontrollers, Analog Circuits, Electric Machines & Motors, and PCB Design.
Format	Concept sessions, guided simulations and hands-on activities using Tinkercad, ESP32, MATLAB and KiCad. Specific project briefs and assessment details are not provided in the supplied schedule.
Support	Participants should follow the setup instructions and material shared by the workshop team. Further resources and support arrangements will be communicated separately.
Certificate	Attendance and certificate requirements, if applicable, will be communicated by the workshop team.

Workshop Objectives

- Build a foundation in electrical circuits, voltage, current, resistance and circuit analysis.
- Understand the role of diodes, transistors and analog circuits in switching and signal conditioning.
- Explore ESP32 microcontrollers and their use in controlling and interfacing with electronic systems.
- Understand the electrical start-up sequence of the buggy and implement a simulated version.
- Learn the operating principles of electric motors and explore motor simulation using MATLAB.
- Understand PCB construction and gain introductory experience designing a board using KiCad.

Workshop note: This document is prepared from the Powertrain schedule supplied by the team. Where a date, venue, speaker, timing or activity was not specified, it is marked as to be notified rather than assumed.

1. Colour Legend

Category	Meaning
Network Analysis	Electrical circuit fundamentals
Electronic Devices	Diodes, transistors and switching circuits
Microcontrollers	ESP32 and embedded control
Analog Circuits	Buggy start-up, converters, op-amps and filters
Electric Machines & Motors	Motor principles and MATLAB simulation
PCB Design	PCB fundamentals and KiCad design
Hands-on	Practical simulation and implementation sessions
Online	Sessions delivered online

2. Participant Guidelines

- **Check the venue.** The regular venue is E6 – Mechanical Seminar Hall. The sessions on 14 and 15 November are assigned to E8 – Robotics Seminar Hall.
- **Join online sessions.** The 17 October and 28 October sessions are marked online; joining details will be shared by the workshop team.
- **Be prepared.** Bring a laptop for simulation and design activities, particularly the Tinkercad, MATLAB and KiCad sessions. Any specific hardware requirements will be communicated separately.
- **Complete the setup.** Install or access the required tools before the relevant hands-on sessions, following the instructions provided by the team.
- **Participate in the hands-on work.** Follow the demonstrations, build the circuits or simulations, and ask questions when a connection or result is unclear.
- **Keep the schedule handy.** Dates, speakers and delivery modes are reproduced from the supplied schedule. Any changes will be communicated by the workshop team.
- **Further details.** Session timings, the purpose and content of 14–15 November, and any certificate or attendance requirements are to be notified.

LAPTOP / TOOLS

A laptop is recommended for the simulation and PCB-design sessions. Tools explicitly named in the schedule include Tinkercad, ESP32, MATLAB and KiCad. Exact versions and installation instructions are to be shared by the workshop team.

3. Timetable

The timetable below follows the dates and presenters in the supplied Powertrain schedule. Session timings have not been specified.

No.	Date	Day	Subject	Session / Topic	Presented by	Venue
1	7 Oct 2026	Wednesday	Network Analysis	Circuit Basics: How Electricity Flows	Kamaleshvar (Alumnus) & Akshayaa	E6
2	9 Oct 2026	Friday	Electronic Devices	Diodes & Transistors: The Tiny Switches	Dillimaran (Alumnus) & Kathin	E6
3	12 Oct 2026	Monday	Microcontrollers	Meet the Microcontroller: ESP32 Basics	Dr. Sujin (Dept. of ECE)	E6
4	14 Oct 2026	Wednesday	Electronic Devices	Hands-on 1: Simulating a Transistor Switch	Team	E6
5	17 Oct 2026	Saturday	Analog Circuits	How Our Buggy Starts Up (Online)	Allwin (Alumnus) & Akshayaa	Online
6	21 Oct 2026	Wednesday	Analog Circuits / Microcontrollers	Hands-on 2: Buggy Start-Up in Tinkercad and ESP32 – Day 1	Team	E6
7	23 Oct 2026	Friday	Analog Circuits / Microcontrollers	Hands-on 3: Buggy Start-Up in Tinkercad and ESP32 – Day 2	Team	E6
8	26 Oct 2026	Monday	Electric Machines & Motors	How Electric Motors Work	Aravind	E6
9	28 Oct 2026	Wednesday	Analog Circuits	Power Up & Clean Up: Buck-Boost Converters, Op-Amps & Filters (Online)	Priyanka	Online
10	4 Nov 2026	Wednesday	PCB Design	What Is a PCB and How Is It Made?	Nanthavikraman (Alumnus)	E6
11	11 Nov 2026	Wednesday	Electric Machines & Motors	Hands-on 4: Simulating Electric Motors in MATLAB	Dr. Ravikrishna (Dept. of EEE)	E6
12	12 Nov 2026	Thursday	PCB Design	Hands-on 5: Laying Out a Board and Designing Your First PCB Using KiCad	Saswin (Alumnus) & Nanthavikraman (Alumnus)	E6

Additional venue dates

Date	Venue	Session details
14 Nov 2026	E8 – Robotics Seminar Hall	Topic, presenter and timing to be notified
15 Nov 2026	E8 – Robotics Seminar Hall	Topic, presenter and timing to be notified

SESSION 1 · NETWORK ANALYSIS

Circuit Basics: How Electricity Flows

Date Wednesday, 7 October 2026
Subject NETWORK ANALYSIS
Presented by Kamaleshvar (Alumnus) & Akshayaa
Venue E6 – Mechanical Seminar Hall

Before working with electronic devices, it is important to understand how electrical quantities behave in a circuit. This session introduces the basic ideas needed to read and analyse simple electrical networks.

What you will learn

- Understand voltage, current, resistance and electrical power.
- Apply Ohm's Law to simple circuit calculations.
- Differentiate series and parallel circuit connections.
- Read and interpret a basic circuit diagram.

Topics covered

01 Electrical quantities
Voltage, current, resistance and power, and the role each plays in describing a circuit.

02 Ohm's Law
Relationship between voltage, current and resistance, with simple numerical examples.

03 Series and parallel circuits
How current and voltage behave in different circuit arrangements.

04 Circuit analysis basics
Recognising components, connections and paths in a simple schematic.

Tools & tech stack

Circuit diagrams and basic electrical-analysis material. Specific software or equipment is to be notified.

Before you arrive

No prerequisites were specified in the supplied schedule.

SESSION 2 · ELECTRONIC DEVICES

Diodes & Transistors: The Tiny Switches

Date Friday, 9 October 2026
Subject ELECTRONIC DEVICES
Presented by Dillimaran (Alumnus) & Kathin
Venue E6 – Mechanical Seminar Hall

Diodes and transistors are building blocks of electronic circuits. This session introduces their behaviour and explains how they are used for rectification, switching and control.

What you will learn

- Understand the basic operation of a diode and a transistor.
- Recognise common diode and transistor circuit symbols.
- Explain how a transistor can operate as an electronic switch.
- Identify practical uses of these devices in electronic circuits.

Topics covered

01 Diode fundamentals
Polarity, forward and reverse bias, and the basic current-voltage behaviour of a diode.

02 Diode applications
Common uses of diodes in electronic circuits, including rectification and protection.

03 Transistor fundamentals
The basic terminals and operating idea of a transistor.

04 Transistor as a switch
How a control signal can switch a load between OFF and ON states.

Tools & tech stack

Electronic-device diagrams and demonstration material. Specific components or simulation tools are to be notified.

Before you arrive

Review basic voltage, current and resistance.

SESSION 3 · MICROCONTROLLERS

Meet the Microcontroller: ESP32 Basics

Date Monday, 12 October 2026
Subject MICROCONTROLLERS
Presented by Dr. Sujin (Dept. of ECE)
Venue E6 – Mechanical Seminar Hall

Microcontrollers provide a compact way to read inputs and control outputs. This session introduces the ESP32 and its role in embedded systems.

What you will learn

- Understand what a microcontroller does in an embedded system.
- Identify the basic features and interfaces of an ESP32 board.
- Understand GPIO input and output at an introductory level.
- Recognise how a microcontroller can interface with sensors and actuators.

Topics covered

01 Microcontroller overview
The role of a microcontroller and how it differs from a general-purpose computer.

02 ESP32 introduction
An overview of the ESP32 development board and its basic capabilities.

03 GPIO basics
Using general-purpose input/output pins to read inputs and control outputs.

04 Embedded applications
Examples of microcontrollers interacting with sensors and electronic components.

Tools & tech stack

ESP32 development board and associated programming environment; exact setup details are to be notified.

Before you arrive

No prior ESP32 experience was specified as a requirement.

SESSION 4 · ELECTRONIC DEVICES · HANDS-ON

Hands-on 1: Simulating a Transistor Switch

Date Wednesday, 14 October 2026
Subject ELECTRONIC DEVICES · HANDS-ON
Presented by Handled jointly by the team
Venue E6 – Mechanical Seminar Hall

This practical session applies the transistor concepts introduced earlier. Participants will simulate a transistor switching circuit and observe how a control input changes the state of a load.

What you will learn

- Translate a simple transistor-switch schematic into a simulated circuit.
- Identify the connections required for a transistor switching circuit.
- Observe the ON and OFF behaviour of a simulated switch.
- Relate the simulation to practical electronic load control.

Topics covered

01 Circuit setup
Identify the transistor, supply, load and control input in the circuit.

02 Simulation
Build the circuit in the selected simulator and check the connections.

03 Switching behaviour
Change the input condition and observe the load response.

04 Troubleshooting
Check wiring and component orientation when the expected behaviour is not observed.

Tools & tech stack

Circuit simulation platform; the exact platform is to be confirmed by the team.

Before you arrive

Review the diode and transistor fundamentals from the 9 October session.

SESSION 5 · ANALOG CIRCUITS

How Our Buggy Starts Up

Date Saturday, 17 October 2026
Subject ANALOG CIRCUITS
Presented by Allwin (Alumnus) & Akshayaa
Venue Online

A buggy's start-up depends on a sequence of electrical and electronic actions. This online session introduces the start-up process and the circuits involved.

What you will learn

- Understand the purpose of the buggy start-up circuit.
- Trace the basic flow of power through the start-up system.
- Recognise the role of switching and control components.
- Prepare for the practical start-up implementation sessions.

Topics covered

01 Start-up system overview
An introductory look at the electrical system involved in starting the buggy.

02 Power flow
How electrical power is supplied and routed through the start-up circuit.

03 Switching and control
The role of switches, relays or other control elements where used in the buggy system.

04 Start-up sequence
A step-by-step view of the sequence that leads to the buggy starting.

Tools & tech stack

Online presentation and circuit diagrams. Joining details will be shared by the workshop team.

Before you arrive

Review basic circuit concepts and transistor switching.

SESSION 6 · ANALOG CIRCUITS / MICROCONTROLLERS · HANDS-ON

Hands-on 2: Build the Buggy's Start-Up – Day 1

Date Wednesday, 21 October 2026
Subject ANALOG CIRCUITS / MICROCONTROLLERS · HANDS-ON
Presented by Handled jointly by the team
Venue E6 – Mechanical Seminar Hall

The first day of the practical build translates the buggy start-up concept into a circuit simulation and an ESP32-based implementation.

What you will learn

- Break the start-up process into circuit-level steps.
- Build the initial start-up circuit in Tinkercad.
- Identify where ESP32 control fits into the implementation.
- Test individual stages before integrating the complete sequence.

Topics covered

01 Translate the start-up sequence
Convert the sequence discussed in the online session into a circuit plan.

02 Tinkercad implementation
Place and connect the required components in a Tinkercad circuit.

03 ESP32 setup
Introduce the ESP32 into the control arrangement and identify relevant connections.

04 Initial testing
Test the circuit stages and identify issues for the next hands-on day.

Tools & tech stack

Tinkercad and ESP32. Exact component list and software setup are to be shared by the team.

Before you arrive

Attend or review the 17 October session on the buggy start-up system.

SESSION 7 · ANALOG CIRCUITS / MICROCONTROLLERS · HANDS-ON

Hands-on 3: Build the Buggy's Start-Up – Day 2

Date Friday, 23 October 2026
Subject ANALOG CIRCUITS / MICROCONTROLLERS · HANDS-ON
Presented by Handled jointly by the team
Venue E6 – Mechanical Seminar Hall

The second day continues the start-up implementation. Participants build on the previous day's circuit, test the control sequence and troubleshoot the integrated simulation.

What you will learn

- Continue and complete the start-up circuit implementation.
- Connect the ESP32 control logic with the simulated circuit.
- Test the expected start-up sequence.
- Troubleshoot wiring and control-flow issues.

Topics covered

01 Continue the circuit build
Review the Day 1 implementation and complete remaining connections.

02 ESP32 control
Develop or complete the ESP32 control required for the start-up sequence.

03 Integrated testing
Run the complete sequence and observe whether the expected output is achieved.

04 Debugging and review
Identify common circuit and logic issues and review the final implementation.

Tools & tech stack

Tinkercad and ESP32. Exact component list and software setup are to be shared by the team.

Before you arrive

Bring the Day 1 circuit and notes from the 21 October session.

SESSION 8 · ELECTRIC MACHINES & MOTORS

How Electric Motors Work

Date Monday, 26 October 2026
Subject ELECTRIC MACHINES & MOTORS
Presented by Aravind
Venue E6 – Mechanical Seminar Hall

Electric motors convert electrical energy into mechanical motion. This session introduces the principles behind motor operation and their relevance to electric and automotive systems.

What you will learn

- Explain the basic principle of electric motor operation.
- Understand the role of magnetic fields in producing motion.
- Recognise key motor characteristics such as speed and torque.
- Identify the relevance of electric motors in vehicle applications.

Topics covered

01 Motor fundamentals
An introduction to the conversion of electrical energy into mechanical energy.

02 Electromagnetic principles
How magnetic fields and current interact to produce force and rotation.

03 Speed and torque
An introductory look at two important motor performance parameters.

04 Automotive applications
Where electric motors are used in vehicle systems.

Tools & tech stack

Presentation material and motor diagrams or demonstrations, as arranged by the presenter.

Before you arrive

No prerequisites were specified in the supplied schedule.

SESSION 9 · ANALOG CIRCUITS

Power Up & Clean Up: Buck-Boost Converters, Op-Amps & Filters

Date Wednesday, 28 October 2026
Subject ANALOG CIRCUITS
Presented by Priyanka
Venue Online

Power converters regulate voltage levels, operational amplifiers support analog signal processing, and filters help condition signals. This online session introduces these important analog-circuit building blocks.

What you will learn

- Understand the purpose of buck and boost conversion.
- Recognise the role of an operational amplifier in analog circuits.
- Explain why filters are used to condition signals.
- Relate power regulation and signal conditioning to electronic systems.

Topics covered

01 Buck and boost converters
The basic purpose of stepping DC voltage down or up.

02 Buck-boost conversion
An introduction to converter arrangements that can provide step-up or step-down operation.

03 Operational amplifiers
Basic op-amp behaviour and their use in analog signal circuits.

04 Filters
Why signals are filtered and how filters help reduce unwanted frequency components.

Tools & tech stack

Online presentation and circuit diagrams. Joining details will be shared by the workshop team.

Before you arrive

Review basic circuit quantities and analog-circuit concepts.

SESSION 10 · PCB DESIGN

What Is a PCB and How Is It Made?

Date Wednesday, 4 November 2026
Subject PCB DESIGN
Presented by Nanthavikraman (Alumnus)
Venue E6 – Mechanical Seminar Hall

A printed circuit board provides the physical platform that connects electronic components. This session introduces PCB structure, design concepts and the broad manufacturing process.

What you will learn

- Understand the purpose and structure of a printed circuit board.
- Recognise common PCB layers and features.
- Understand how a schematic becomes a physical board layout.
- Get an overview of the PCB manufacturing process.

Topics covered

01 PCB fundamentals
What a PCB is and why it is used instead of loose point-to-point wiring.

02 Board structure
An introduction to substrates, copper layers, tracks, pads and vias.

03 From schematic to layout
How circuit connections are represented in a board design.

04 Manufacturing overview
A high-level look at how a designed PCB is fabricated and assembled.

Tools & tech stack

PCB examples, design references and presentation material. KiCad is used in the later hands-on session.

Before you arrive

No prerequisites were specified in the supplied schedule.

SESSION 11 · ELECTRIC MACHINES & MOTORS · HANDS-ON

Hands-on 4: Simulating Electric Motors in MATLAB

Date Wednesday, 11 November 2026
Subject ELECTRIC MACHINES & MOTORS · HANDS-ON
Presented by Dr. Ravikrishna (Dept. of EEE)
Venue E6 – Mechanical Seminar Hall

This practical session introduces motor simulation using MATLAB. Participants explore how a motor model can be used to study its behaviour and performance.

What you will learn

- Understand the purpose of modelling an electric motor.
- Use MATLAB as a tool for motor simulation.
- Observe motor response through simulation outputs and plots.
- Relate simulation results to motor performance concepts.

Topics covered

01 Motor model overview
Understand the purpose and basic inputs and outputs of a motor model.

02 MATLAB environment
Get oriented with the MATLAB workspace and simulation workflow.

03 Running a simulation
Execute a motor simulation using the model and parameters provided.

04 Reading results
Interpret the plots or outputs produced by the simulation.

Tools & tech stack

MATLAB. Required version, files and setup instructions are to be shared by the presenter.

Before you arrive

Review the electric motor fundamentals from the 26 October session.

SESSION 12 · PCB DESIGN · HANDS-ON

Hands-on 5: Laying Out a Board and Designing Your First PCB Using KiCad

Date Thursday, 12 November 2026
Subject PCB DESIGN · HANDS-ON
Presented by Saswin (Alumnus) & Nanthavikraman (Alumnus)
Venue E6 – Mechanical Seminar Hall

This hands-on session takes participants through the introductory PCB-design workflow in KiCad, from schematic capture to board layout.

What you will learn

- Navigate the basic KiCad design workflow.
- Create or work with a simple circuit schematic.
- Assign footprints and place components on a board.
- Route tracks and inspect the resulting PCB layout.

Topics covered

01 Project and schematic
Create a KiCad project and prepare the circuit schematic.

02 Footprints and placement
Associate components with footprints and arrange them on the board.

03 Track routing
Connect pads using copper tracks while keeping the layout organised.

04 Design review
Inspect the completed layout and understand the next steps before fabrication.

Tools & tech stack

KiCad. Installation instructions and any starter design files are to be shared by the team.

Before you arrive

Review the PCB fundamentals session from 4 November and bring a laptop with KiCad ready.

4. Subject-wise Summary

Subject	Presented by	Sessions	Dates
Network Analysis	Kamaleshvar (Alumnus) & Akshayaa	1	7 Oct 2026
Electronic Devices	Dillimaran (Alumnus) & Kathin; team	2	9, 14 Oct 2026
Microcontrollers	Dr. Sujin (Dept. of ECE)	1	12 Oct 2026
Analog Circuits	Allwin (Alumnus) & Akshayaa; Priyanka; team	4	17, 21, 23, 28 Oct 2026
Electric Machines & Motors	Aravind; Dr. Ravikrishna (Dept. of EEE)	2	26 Oct, 11 Nov 2026
PCB Design	Nanthavikraman (Alumnus); Saswin (Alumnus) & Nanthavikraman (Alumnus)	2	4, 12 Nov 2026

Session Count

12 scheduled sessions from 7 October to 12 November 2026. The 14 and 15 November venue dates are listed separately because their session topics, presenters and timings were not included in the supplied schedule.

Delivery Mode

The sessions on 17 October and 28 October are explicitly marked online. The remaining dated sessions are assigned to E6 – Mechanical Seminar Hall. The 14 and 15 November dates are assigned to E8 – Robotics Seminar Hall.

5. Hands-on Sessions and Practical Work

The supplied schedule identifies five hands-on sessions. The following table records the practical activity stated for each session without adding separate project requirements.

Activity	Date	Practical work	Tool / platform	Handled by
Hands-on 1	14 Oct 2026	Simulating a Transistor Switch	Circuit simulation	Team
Hands-on 2	21 Oct 2026	Build the Buggy's Start-Up in Tinkercad and using ESP32 – Day 1	Tinkercad / ESP32	Team
Hands-on 3	23 Oct 2026	Build the Buggy's Start-Up in Tinkercad and using ESP32 – Day 2	Tinkercad / ESP32	Team
Hands-on 4	11 Nov 2026	Simulating Electric Motors in MATLAB	MATLAB	Dr. Ravikrishna
Hands-on 5	12 Nov 2026	Design Your First PCB using KiCad	KiCad	Saswin & Nanthavikraman

Preparation checklist

- Review the relevant concept session before attending its hands-on continuation.
- Bring a laptop for Tinkercad, MATLAB and KiCad activities.
- Follow the setup and component requirements circulated by the workshop team.
- Save your simulation files and design work so that you can continue or review them later.
- Ask the team to clarify any required software version, account, component or starter file before the session.

Projects and assessment

The Powertrain schedule supplied for this document does not specify separate mini-projects, larger projects, deadlines or assessment criteria. Any such activities will be included once confirmed by the workshop team.

6. Student Timetable

Powertrain Workshop 2026 · Team ASTERIX · aBAJA SAEINDIA

The calendar below is a date-wise overview. Session timings are to be notified.

Week 1 · 5–11 Oct

Date	Session / activity	Venue / mode
7 Wed	S1 · Network Analysis	E6
9 Fri	S2 · Electronic Devices	E6

Week 2 · 12–18 Oct

Date	Session / activity	Venue / mode
12 Mon	S3 · ESP32 Basics	E6
14 Wed	S4 · Transistor Switch	E6
17 Sat	S5 · Buggy Start-Up	Online

Week 3 · 19–25 Oct

Date	Session / activity	Venue / mode
21 Wed	S6 · Start-Up Build Day 1	E6
23 Fri	S7 · Start-Up Build Day 2	E6

Week 4 · 26 Oct–1 Nov

Date	Session / activity	Venue / mode
26 Mon	S8 · Electric Motors	E6
28 Wed	S9 · Converters, Op-Amps & Filters	Online

Week 5 · 2–8 Nov

Date	Session / activity	Venue / mode
4 Wed	S10 · PCB Fundamentals	E6

Week 6 · 9–15 Nov

Date	Session / activity	Venue / mode
11 Wed	S11 · Motor Simulation (MATLAB)	E6
12 Thu	S12 · PCB Design (KiCad)	E6
14 Sat	Venue: E8 Robotics Seminar Hall	Details TBA
15 Sun	Venue: E8 Robotics Seminar Hall	Details TBA

7. Learning Support and Further Information

- Venue: E6 – Mechanical Seminar Hall for all scheduled in-person sessions, except 14 and 15 November at E8 – Robotics Seminar Hall.
- Online sessions: 17 October and 28 October. Joining links and instructions are to be shared separately.
- Tools: Tinkercad, ESP32, MATLAB and KiCad are named in the schedule. Installation guidance and exact versions are to be notified.

- Unspecified details: timings and the session content, presenters and schedule for 14–15 November are not provided in the source schedule.

Schedule note: The dates, subjects, presenters and online labels in this document are based on the information supplied by the Powertrain workshop team. Any additional details or changes should be communicated through the official workshop updates.